

The Variety of Configurations of Static Liquid Crystals

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Here we make several observations on a variety of special classes of harmonic maps from domains in $\mathbb{R}^3$ to $\mathbb{S}^2$. Such maps are relevant for the study of liquid crystals; see e.g. [HK₁], [HKL₁], [HKLu]. Part of our work is motivated by questions about the *nonuniqueness* and the *number* of harmonic maps having fixed boundary data. Here we show that the number may actually be infinite. For example, by Corollary 3.2 below *there exists a one parameter family of distinct energy-minimizers each having the same Dirichlet boundary data*.

In [HK₂, §3] we considered the subclass of *planar harmonic maps*. These are ones that have values in $\mathbb{S}^1 = \mathbb{S}^2 \cap \{z = 0\}$. With M. Luskin, we found an elementary example of a smooth planar harmonic map on a cube which is not energy-minimizing when considered as a map into $\mathbb{S}^2$. Thus, an energy-minimizer provided another stationary harmonic map with the same boundary values. In §1, we check that this energy minimizer $u = (u_1, u_2, u_3)$ is itself nonplanar. Hence, $(u_1, u_2, |u_3|)$ and $(u_1, u_2, -|u_3|)$ are both minimizers that differ from the original planar stationary harmonic map. Showing that $u_3 \not\equiv 0$ involves verifying the interesting fact that *on a connected region, there exists no nonconstant H^1 function whose trace assumes only a discrete set of values*. The above reflection symmetry was used in [SU₃] and occurs in Almgren and Lieb's [AL] example of nonuniqueness, which is based on the behavior of singularities. In another example, Hardt and Lin [HL₃, §5] also proved the existence of boundary data for which one minimizer has singularities while another minimizer does not.

Here our one parameter family of minimizers is constructed by breaking a putative *axial symmetry*. Here, for each positive integer n , a map u is called *n-axially symmetric* if, in cylindrical coordinates,

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$$u(r, \theta, z) = (\cos n\theta \sin \varphi, \sin n\theta \sin \varphi, \cos \varphi)$$

for some real-valued function $\varphi(r, z)$. This implies that $u \circ R_\theta \equiv R_{n\theta} \circ u$ for any $\theta \in [0, 2\pi]$ where R_θ denotes rotation about the z -axis through an angle θ . We show that, *in spite of n -axially symmetric boundary data, an energy-minimizer u with singularities cannot be n -axially symmetric when $n \geq 2$* . In fact here the equation $u \circ R_\theta \equiv R_{n\theta} \circ u$ can hold for at most finitely many θ . So $\{R_{-n\theta} \circ u \circ R_\theta : -\varepsilon \leq \theta \leq \varepsilon\}$, for ε sufficiently small, is a family of distinct minimizers sharing the same boundary values. Energy minimizers must necessarily have singularities if the given boundary data has nonzero degree or is sufficiently oscillatory [HL₁]. In retrospect, the first indication of this symmetry breaking was in the numerical work of M. Luskin, S.Y. Lin, and R. Cohen [CHKLL]. Their results indicated that the specific 2-axially symmetric boundary data corresponding to the rational functional z^2 would give a minimizer having 2 off-axis singularities of degree 1. Concerning 1-axial symmetry, we also find in 5.2 certain 1-axially symmetric boundary data for which a minimizer cannot be 1-axially symmetric. For the general liquid crystal functional, F. Helein [H], and independently S.Y. Lin [Li], verified that the critical point $x/|x|$ is not a minimizing configuration for various ranges of constants. Using this, J. Ericksen c.f. [K] observed the necessity of rotational symmetry breaking and nonuniqueness for such liquid crystals. Here there is a two-parameter family of minimizers sharing the same boundary data.

The pure axially symmetric problem in which one considers critical points in the class of axially symmetric harmonic maps has been studied by D. Zhang [Z]. He obtained a smooth axially symmetric harmonic map corresponding to any given smooth axially symmetric boundary data that omits a neighborhood of the South Pole. §3 contains a slightly improved version of this result (perhaps optimal) in which one allows boundary data that can reach the South Pole but not wrap around. For a mapping that minimizes energy among axially symmetric maps, the singularities must be isolated (4.1). Also the singularities for such minimizers are classified, and a minimizer converges asymptotically to a unique tangent map on approach to a singular point (4.2). Examples with a prescribed number of singular points are given in §5. Some other constructions controlling the location and number of singularities in minimizing harmonic maps have been given in [HL₁] and [AL]. It is also observed in §6 that minimizers in the axially symmetric subclass, unlike absolute minimizers, do not satisfy the “Caccioppoli-type” estimate of [HL₂, 6.2] c.f. also [HKL₁], [HKL₂].

Harmonic maps between spheres with specified symmetries have been employed by B. Smith [S] to represent homotopy groups of spheres. As was observed recently by Ding [D], these may be constructed by minimiz-

ing energy among classes of maps with prescribed symmetries. W. Jager and H. Kaul [JK₂] and A. Baldes [B] have discussed rotational symmetry for harmonic maps to spheres and ellipsoids. See [CH] for references and more recent results on rotational symmetry. It is interesting to compare symmetry results of harmonic maps with those of minimal surfaces [M₂]. F. Morgan [M₁] has constructed in $\mathbb{R}^3$ a one parameter family of distinct minimal surfaces each having the same four circles as boundary. However, these minimal surfaces are necessarily unstable, and it is known that, for a boundary in the sphere invariant under a connected Lie group of symmetries, an area-minimizing hypersurface, even with singularities, inherits the symmetries of its boundary [La].

For recent developments in liquid crystal theory, see [LFH], [K], [Ha], and other paper in this volume.

1. Planar Harmonic Maps

Suppose Ω is a bounded domain in $\mathbb{R}^3$ and $v \in H^{1/2}(\partial\Omega, \mathbb{S}^1)$. Recall from [HL₂, 6.3] that there may exist no mapping $u \in H^1(\Omega, \mathbb{S}^1)$ whose trace on $\partial\Omega$ is v . However, if there is such a finite energy extension, then the resulting least-energy u is unique by the following remark:

For any $v \in H^{1/2}(\partial\Omega, \mathbb{S}^1)$, there exists at most one harmonic map $u \in H^1(\Omega, \mathbb{S}^1)$ with $u|_{\partial\Omega} = v$.

To see this, recall [HK₂, §3] that a harmonic mapping $u \in H^1(\Omega, \mathbb{S}^1)$ has the form $(\cos \theta, \sin \theta, 0)$ where θ is a smooth harmonic function on Ω . Suppose u_1 and u_2 are two such harmonic mappings with $u_1|_{\partial\Omega} = u_2|_{\partial\Omega}$. Then, for the corresponding harmonic functions θ_1 and θ_2 , the trace of $\theta_1 - \theta_2$ on $\partial\Omega$ has values in the discrete set $\{2\pi j: j \in \mathbb{Z}\}$. Thus the above remark follows from the:

1.1 Lemma. *On a Lipschitz domain $\Omega \subset \mathbb{R}^m$, there exists no non-constant H^1 function v whose trace assumes only a discrete set of values.*

Proof. The lemma is local in character so we may suppose that a portion of Ω contains the half ball $G = \{x: x_m > 0, |x| < 1\}$ and that $\partial\Omega \cap \bar{G} = \Gamma = \{x_m = 0, |x'| < 1\}$. Replacing v by $\psi \circ v$ for a suitable smooth function ψ , we may assume that $v|_{\Gamma}$ equals 0 or 1. Then in case $m = 2$, the comparison of the $H^{1/2}$ norm of $v|_{\Gamma}$ with the Marcinkiewicz integral [St, p. 14] easily convinces the reader that $v|_{\Gamma}$ is constant. Suppose

inductively that the Lemma is true in dimensions $\leq m - 1$. Since

$$\int_G |\nabla v|^2 dx < \infty,$$

there is, by Fubini's theorem, for almost every $a \in (-1, 1)$, an $m - 1$ dimensional domain

$$D(a) = G \cap \{x_1 = a\}$$

so that $\int_{D(a)} |\nabla v|^2 dx_2 \cdots dx_m < \infty$. Moreover, $v|_{\Gamma \cap \{x_1 = a\}}$ equals 0 or 1. By induction $v|_{D(a)}$ is constant. It is easy to check that the constant is independent of the choice of a . $\square$

An old version of this fact, due to Wiener [C] states that a harmonic function in two variables with finite Dirichlet integral cannot suffer jump discontinuities on the boundary of its domain of definition.

In [HK₂, §3], it was shown that, *for sufficiently large q , the harmonic mapping*

$$n_q(x, y, z) = (\cos qz, \sin qz, 0)$$

on the unit cube Q in $\mathbb{R}^3$, is not energy minimizing as a map into S^2 . Choosing $u = (u_1, u_2, u_3)$ to minimize energy among maps in $H^1(Q, S^2)$ with $u|_{\partial Q} = n_q|_{\partial Q}$, we conclude from the above remark that $u_3 \not\equiv 0$. Thus There are at least three harmonic maps in $H^1(Q, S^2)$ having trace $n_q|_{\partial Q}$; these include n_q and the energy-minimizers $(u_1, u_2, |u_3|)$ and $(u_1, u_2, -|u_3|)$.

2. A Continuum of Minimizers

Let $(r, \theta, z) \in \mathbb{R}^+ \times [0, 2\pi] \times \mathbb{R}$ be the standard cylindrical coordinates of $\mathbb{R}^3$, and let n be a positive integer. A map u from a subset of $\mathbb{R}^3$ into S^2 is called *n -axially symmetric* if $u = u_\varphi$ where

$$(2.1) \quad u_\varphi(r, \theta, z) = (\cos n\theta \sin \varphi, \sin n\theta \sin \varphi, \cos \varphi)$$

for some real-valued function $\varphi(r, z)$. This implies that

$$u \circ R_\theta = R_{n\theta} \circ u$$

for any $\theta \in [0, 2\pi]$ where R_θ denotes rotation about the z -axis through an angle θ . Note that an n -axially symmetric u will be continuous at a point $(0, 0, z)$ on the Z -axis only if $\varphi(0, 0, z)$ equals an integer multiple of

π . Given φ , we will call u_φ the *n-axially symmetric extension* of φ . For example, if

$$\varphi = \arccot \left(r / [\sqrt{r^2 + z^2} - z] \right)^n$$

on the half-disk $\{(r, z): 0 \leq r^2 \leq 1 - z^2\}$, then u_φ is the homogeneous-degree-0 extension of the conformal map of S^2 given in stereographic projection by z^n .

2.1 Theorem. *Any energy minimizing map $u: \mathbb{B}^3 \rightarrow S^2$ with singularities is not n-axially symmetric for $n \geq 2$. In fact, there are at most a finite number of angles $\theta \in [0, 2\pi)$ such that $u \circ R_\theta \equiv R_{n\theta} \circ u$.*

Proof. By $[SU_1]$ the set of singularities of u is a discrete subset of $\mathbb{B} = \mathbb{B}^3$ characterized by density condition

$$(2.2) \quad \text{Sing } u = \left\{ a \in \mathbb{B}: \limsup_{r \rightarrow 0} r^{-1} \int_{\mathbb{B}_{r(a)}} |\nabla u|^2 dx > 0 \right\}.$$

First suppose that a is a singular point of u lying on the Z -axis and that $u_a(x) = \lim_{i \rightarrow \infty} u(\lambda_i(x - a))$ is a tangent map of u of a . Since this convergence is pointwise a.e., we would obtain the relation $u_a \circ R_\theta \equiv R_{n\theta} \circ u_a$ whenever $u \circ R_\theta \equiv R_{n\theta} \circ u$ was true. But the classification of $[BCL]$ implies that such a tangent map must have the form $u_a(x) = g_a(x/|x|)$ for some rotation g_a of $\mathbb{R}^3$, from which we deduce that $u_a \circ R_\theta \not\equiv R_{n\theta} \circ u_a$ for every $\theta \in (0, 2\pi)$.

It remains to consider the case where there is a singular point a of u lying off the Z -axis. From (2.2) it is clear that each distinct angle $\theta \in [0, 2\pi)$ satisfying $u \circ R_\theta \equiv R_{n\theta} \circ u$, then gives rise to a distinct point $R_\theta(a)$ in the intersection of $\text{Sing } u$ with the horizontal circle $\{x_1^2 + x_2^2 = a_1^2 + a_2^2, x_3 = a_3\}$. Thus the number of such θ is finite. $\square$

2.2 Corollary. *For any nonsmooth energy minimizing map $u: \mathbb{B}^3 \rightarrow S^2$ whose trace on $\partial\mathbb{B}^3$ is n-axially symmetric with $n \geq 2$,*

$$\{R_{-n\theta} \circ u \circ R_\theta: -\varepsilon \leq \theta \leq \varepsilon\}$$

forms, for ε sufficiently small, a one parameter family of distinct energy-minimizing maps each having the same Dirichlet boundary data.

3. Regular Axially Symmetric Harmonic Maps

For the remainder of the paper, we will consider critical points of the energy functional on axially symmetric maps. For simplicity, axial symmetry will mean 1-axial symmetry. All of the results, with a few changes of formulas, continue to hold for n -axial symmetry. See 6.3.

For a real-valued function φ on the half-disk $D = \{(r, z): 0 \leq r^2 \leq 1 - z^2\}$, one readily computes the energy of the axially symmetric extension of φ ,

$$\mathbf{E}(\varphi) = \int_{\mathbf{B}} |\nabla u_\varphi|^2 dx = \int_D [r(\partial\varphi/\partial r)^2 + r(\partial\varphi/\partial z)^2 + r^{-1} \sin^2 \varphi] dr dz.$$

Thus φ is a critical point for $\mathbf{E}$ if it satisfies the equation

$$(3.1) \quad \partial/\partial r(r\partial\varphi/\partial r) + \partial/\partial z(r\partial\varphi/\partial z) + \frac{1}{2}r^{-1} \sin 2\varphi = 0 \text{ in } D.$$

Dong Zhang has proven [Z] that, for any $\delta > 0$, equation (3.1) with the boundary condition

$$(3.2) \quad \varphi = g \text{ on } \partial D$$

has a solution $\varphi \in C^0(\overline{D}) \cap C^\infty(D)$ for any Lipschitz function

$$g: \partial D \rightarrow [0, \pi - \delta] \text{ with } g(0, z) \equiv 0.$$

Note that the axially symmetric extension u_φ of any $C^0(\overline{D}) \cap C^\infty(D)$ solution φ of (3.1) is a continuous, hence real analytic, harmonic map from $\mathbf{B}$ to $\mathbf{S}^2$. Thus we describe such a φ as a *regular solution*.

3.1 Theorem. *Problem (3.1)–(3.2) has a regular solution φ for any Lipschitz function*

$$g: \partial D \rightarrow [-\pi, \pi] \text{ with } g(0, z) \equiv 0.$$

We follow closely the argument of [Z] and use the additional:

3.2 Lemma. *Suppose $0 < \eta < \frac{1}{2}$, $0 < \delta < 1 - 2\eta$, and $\Omega_\delta = D \cap \{r > \delta\}$. For any Lipschitz function*

$$g: \partial\Omega_\delta \rightarrow [-\pi, \pi] \text{ with } |g| \leq \pi - \eta \text{ on } \partial\Omega_\delta \cap \{\delta < r < \delta + 2\eta\},$$

an $\mathbf{E}$ -minimizer φ with $\varphi|_{\partial\Omega_\delta} = g$ has

$$|\varphi| \leq \pi - c(\eta) \text{ on } \{r = \delta + \eta\}$$

for some positive $c(\eta)$ depending only on η .

Proof. If the Lemma were false for some positive $\eta < \frac{1}{2}$, then there would exist a sequence $\delta_i \in (0, 1 - 2\eta)$, Lipschitz functions

$$g_i: \partial\Omega_{\delta_i} \rightarrow [-\pi, \pi] \text{ with } |g_i| \leq \pi - \eta \text{ on } \partial\Omega_{\delta_i} \cap \{\delta_i < r < \delta_i + 2\eta\},$$

and E-minimizers φ_i with $\varphi_i|_{\partial\Omega_{\delta_i}} = g_i$ but

$$\|\varphi_i\|_{L^\infty\{r=\delta_i+\eta\}} > \pi - i^{-1}.$$

Note that $\|\varphi_i\|_{L^\infty} \leq \pi$ by the minimality of φ_i . Thus, by elliptic regularity theory, we may assume, by passing to a subsequence, that

$$\begin{aligned} \delta_i &\rightarrow \delta_0 \in [0, 1 - 2\eta], \quad \varphi_i \rightarrow \varphi_0 \text{ in } C_{loc}^2(\Omega_{\delta_0}), \text{ and} \\ \varphi_i &\rightarrow \varphi_0 \text{ uniformly on } \{r = \delta_0 + \eta\}. \end{aligned}$$

Moreover, φ_0 will satisfy (3.1) on Ω_{δ_0} ,

$$\begin{aligned} \|\varphi_0\|_{L^\infty} &\leq \pi, \quad \|\varphi_0\|_{L^\infty\{r=\delta_0+\eta\}} = \pi, \text{ and} \\ |\varphi_0| &\leq \pi - \eta \text{ on } \partial\Omega_{\delta_0} \cap \{\delta_0 < r < \delta_0 + 2\eta\}. \end{aligned}$$

These facts will contradict the Hopf maximum principle. In fact, the function $v = \varphi_0 - \pi$ is nonpositive in Ω_{δ_0} and (as we may assume) vanishes at some point $x_0 \in \Omega_{\delta_0}$. Since v satisfies an elliptic equation of the form

$$a_{ij}(x)(\partial^2/\partial x_i \partial x_j)v + b_i(x)(\partial/\partial x_i)v + c(x)v = 0$$

in a neighborhood of x_0 , $v \equiv 0$ near x_0 . (Note that the sign of $c(x)$ does not play a role here.) By analytic continuation, $\varphi_0 - \pi = v \equiv 0$ in Ω_{δ_0} , contradicting the boundary condition. $\square$

Proof of Theorem 3.1. Since g is Lipschitz continuous on ∂D with $g(0, z) \equiv 0$, we may choose a positive $\eta < \frac{1}{2}$ such that $|g| \leq \pi - 2\eta$ on $\partial D \cap \{r < 3\eta\}$. Modifying [Z], we choose, for a positive $\varepsilon \ll c(\eta)$, functions $\Phi_\varepsilon^\pm = \Phi_\varepsilon^\pm(r)$ satisfying

$$\begin{aligned} d/dr(rd\Phi_\varepsilon^\pm/dr) - \frac{1}{2}r^{-1}\sin 2\Phi_\varepsilon^\pm &= 0 \text{ on } \{0 < r < 2\eta\}, \\ \Phi_\varepsilon^\pm(2\eta) &= \pm(\pi - \varepsilon), \\ \Phi_\varepsilon^\pm &\text{ are smooth with } |\Phi_\varepsilon^\pm| \geq \pi - \varepsilon \text{ on } \{r \geq 2\eta\}, \text{ and} \\ \Phi_\varepsilon^\pm &\equiv \pm\pi \text{ on } \{r \geq 2\eta + \varepsilon\}. \end{aligned}$$

As in [Z], $|\Phi_\varepsilon^\pm| \geq |g|$ on ∂D when ε is sufficiently small, and $\Phi_\varepsilon^\pm(r) \rightarrow \pm\pi$ as $\varepsilon \rightarrow 0$, for all $r > 0$.

Next we let φ be a minimizer of $\mathbf{E}$ subject to the constraints:

$$\varphi \mid \partial D = g \quad \text{and} \quad \Phi_\varepsilon^-(r) \leq \varphi(r, z) \leq \Phi_\varepsilon^+(r).$$

We claim that φ is then a regular solution of (3.1) on all of D . As in [Z], φ is a regular solution on $D \sim \{2\eta \leq r \leq 2\eta + \varepsilon\}$. Moreover, since $|\varphi| \leq \pi$ and $\varphi \mid \partial D = g$, we may apply the maximum principle and the above Lemma (with g replaced by $\varphi \mid \partial D_\delta$, $0 < \delta < 2\eta$) to conclude that φ must satisfy in addition, the inequality $|\varphi| \leq \pi - c(\eta) < \pi - \varepsilon$ on $\{\eta < r < 3\eta\}$. In particular, $\Phi_\varepsilon^- < \varphi < \Phi_\varepsilon^+$, and hence φ satisfies (3.1) also on the middle strip $\{\eta < r < 3\eta\}$. $\square$

4. Minimizers in the Axially Symmetric Class

Suppose $v \in H^{1/2}(\mathbf{S}^2, \mathbf{S}^2)$ is axially symmetric and

$$\mathcal{U}(v) = \{w \in H^1(\mathbf{B}, \mathbf{S}^2) : w \text{ is axially symmetric, } w \mid \partial \mathbf{B} = v\}$$

is nonempty. Note in particular that $v(x/|x|) \in \mathcal{U}(v)$ in case $v \in H^1$. An energy minimizing sequence $\{u_i\}$ in $\mathcal{U}(v)$ contains a subsequence convergent weakly in H^1 to a map $u \in H^1(\mathbf{B}, \mathbf{S}^2)$ with $u \mid \partial \mathbf{B} = v$. Moreover, u is axially symmetric because the convergence of u_i to u is pointwise a.e. By lower semi-continuity, u minimizes energy in $\mathcal{U}(v)$. For the remainder of this section we discuss the regularity of such a *minimizer among axially symmetric maps*.

4.1 Lemma. *There is a positive constant ε_0 such that if u is energy minimizing among axially symmetric maps with $\int_{\mathbf{B}} |\nabla u|^2 dx \leq \varepsilon_0$, then u is smooth on $\mathbf{B}_{1/2}$.*

Proof. As before, $u = u_\varphi$ for a suitable $\varphi: D \rightarrow \mathbb{R}$ and

$$\int_{\mathbf{B}} |\nabla u|^2 dx = \mathbf{E}(\varphi) = \int_D [r(\partial\varphi/\partial r)^2 + r(\partial\varphi/\partial z)^2 + r^{-1} \sin^2 \varphi] dr dz.$$

Since

$$\begin{aligned} |\nabla \cos \varphi| &= |\sin \varphi| \cdot |\partial\varphi/\partial r| + |\sin \varphi| \cdot |\partial\varphi/\partial z| \\ &\leq \frac{1}{2} [r(|\partial\varphi/\partial r|^2 + |\partial\varphi/\partial z|^2) + r^{-1} \sin^2 \varphi], \end{aligned}$$

one has the estimate

$$(4.1) \quad \int_D |\nabla \cos \varphi| dr dz \leq \frac{1}{2} \mathbf{E}(\varphi).$$

For $0 < \rho < 1$, let

$$D_\rho = \{(r, z): 0 \leq r^2 \leq \rho^2 - z^2\} \quad \text{and} \quad \partial^+ D_\rho = \{(r, z): 0 \leq r^2 = \rho^2 - z^2\}.$$

The finiteness of $E(\varphi)$ and Fubini's theorem imply that for almost all ρ , $\varphi|_{\partial^+ D_\rho}$ is continuous with $\varphi(0, \pm\rho) = k_\pm \pi$ for some integer $k_\pm$. By (4.1) and Fubini's theorem, we may also insist that $\frac{1}{2} < \rho < 1$ and

$$\int_{\partial^+ D_\rho} |\nabla \cos \varphi| dr dz \leq E(\varphi).$$

In particular, $\text{osc}_{\partial^+ D_\rho}(\cos \varphi) \leq 2E(\varphi)$, $k_+ = k_-$, and u maps $\partial \mathbb{B}$ continuously into a small neighborhood of the North or South Pole. As in [SU₃] one sees that, by minimality, $u|_{D_\rho}$ must have image in the corresponding open hemisphere. Following [JK₁] and [HKW], $u|_{D_\rho}$ is the unique harmonic map into this hemisphere, with respect to its boundary values, and $u|_{D_\rho}$ is smooth. $\square$

4.2 Theorem. *Suppose $u: \mathbb{B} \rightarrow \mathbb{S}^2$ is energy minimizing among axially symmetric maps. Then $u|_{\mathbb{B}}$ is real analytic away from a set of isolated points on the Z -axis. If $u|_{\partial \mathbb{B}}$ is Lipschitz, then u is also Hölder continuous in a neighborhood of $\partial \mathbb{B}$.*

Proof. We follow many of the arguments used in the partial regularity theory [SU₁] for absolute energy minimizers. By the ellipticity of (3.1) off the Z -axis and by Theorem 4.1,

$$S = \{a \in \mathbb{B}: \limsup_{r \rightarrow 0} r^{-1} \int_{\mathbb{B}_r(a)} |\nabla u|^2 dx > 0\}$$

is a relatively closed subset of $\{r = 0, -1 < z < 1\}$, and $u|_{\mathbb{B}}$ is real analytic off of S . Moreover, S has one dimensional Hausdorff measure zero because $u \in H^1$ [SU₁, 2.7]. To see that S is actually only an isolated set, we first note that homogeneous extension from spheres centered on the Z -axis preserves axial symmetry. So minimality again gives a monotonicity formula as in [SU₁, 2.4] [HL₂, 4.1]. We then may find, for each $a \in S$, a sequence $\lambda_i \rightarrow 0$ so that the maps $u_i = u(\lambda_i(x - a))$ converge weakly in H^1 to an axially symmetric map u_0 that is homogeneous of degree 0, i.e., $u_0(x) = u_0(x/|x|)$. Moreover, since $u_0|_{\mathbb{S}^2} \sim \{(0, 0, -1), (0, 0, 1)\}$ is a harmonic map of finite energy, $u_0|_{\mathbb{S}^2}$ is a smooth harmonic map by [SaU].

At this stage in the argument of [SU₁, 4.6] the convergence of the u_i to u_0 is shown to be *strong* in H^1 . This strong convergence then is used

first to establish the energy minimality of u_0 and second, along with the small-energy-regularity lemma [SU₁, 3.1], to show that singularities are only isolated in dimension 3. With Lemma 4.1 above, these arguments will carry over once we establish this strong convergence.

For this purpose, first note that the energy bounds provided by the monotonicity formula and elliptic regularity theory imply that $u_i \rightarrow u_0$ in $C_{loc}^2(\bar{\mathbf{B}} \sim \{r = 0\})$. For a small positive δ , let $C_\delta^\pm$ denote the solid polar regions

$$C_\delta^\pm = \{x: 1 - \delta < |x| < 1, \ |(x_1, x_2)| < \delta|x|\}.$$

On the shell $\{1 - \delta \leq |x| \leq 1\}$, we use the linear interpolation between u_i and u_0 ,

$$Q_i(x) = \delta^{-1}(|x| + \delta - 1)u_i + \delta^{-1}(1 - |x|)u_0,$$

to define a comparison map

$$\begin{aligned} v_i(x) &= u_0(x) \quad \text{for } |x| \leq 1 - \delta, \\ v_i(x) &= Q_i(x)/|Q_i(x)| \quad \text{for } 1 - \delta \leq |x| \leq 1 \quad \text{and } |(x_1, x_2)| \geq \delta|x|, \\ v_i(x) &= \text{the homogeneous-degree-0 extension of } v_i|_{\partial C_\delta^\pm} \\ &\quad \text{with center } \left(0, 0, \pm \left(1 - \frac{1}{2}\delta\right)\right) \quad \text{for } x \in C_\delta^\pm. \end{aligned}$$

Clearly v_i is axially symmetric with $v_i|_{\partial \mathbf{B}} = u_i$. From the minimality of u_i and the C^2 -convergence in $\bar{\mathbf{B}} \sim C_\delta^\pm$, we find, for i sufficiently large, that

$$\int_{\mathbf{B}} |\nabla u_i|^2 dx \leq \int_{\mathbf{B}} |\nabla v_i|^2 dx \leq \int_{\mathbf{B}_{1-\delta}} |\nabla u_0|^2 dx + c\Lambda\delta$$

where $\Lambda = \sup_{i, 1-\delta \leq \rho \leq 1} \int_{\partial \mathbf{B}_\rho} |\nabla u_i|^2 dx$. By the monotonicity formula, Λ is finite, and we may let $i \rightarrow \infty$ and $\delta \rightarrow 0$ to conclude that

$$\limsup_{i \rightarrow \infty} \int_{\mathbf{B}} |\nabla u_i|^2 dx \leq \int_{\mathbf{B}} |\nabla u_0|^2 dx.$$

Combined with the weak H^1 convergence, this implies the desired strong convergence:

$$\begin{aligned} &\limsup_{i \rightarrow \infty} \int_{\mathbf{B}} |\nabla u_i - \nabla u_0|^2 dx = \\ &\limsup_{i \rightarrow \infty} \int_{\mathbf{B}} |\nabla u_i|^2 dx - \int_{\mathbf{B}} |\nabla u_0|^2 dx - 2 \int_{\mathbf{B}} \langle \nabla u_i - \nabla u_0, \nabla u_0 \rangle dx \leq 0 - 0. \end{aligned}$$

Turning now to the question of boundary regularity for given Lipschitz Dirichlet data, we first note that the ellipticity of (3.1) off the Z -axis leads

readily to the local Hölder continuity of u near points of $\partial\mathbf{B} \sim \{(0, 0, 1), (0, 0, -1)\}$. To deduce Hölder continuity also near the two poles, we modify, as in [SU₂] or [HL₂, §5], the above interior regularity discussion. The continuity of $u|_{\partial\mathbf{B}}$ implies that $u(0, 0, \pm 1) \in \{(0, 0, 1), (0, 0, -1)\}$.

In the boundary version of Lemma 4.1, one assumes, for some positive ρ_0 , that both the oscillation of u on $\partial\mathbf{B} \cap \bar{\mathbf{B}}_{\rho_0}(0, 0, \pm 1)$ and the normalized energy

$$\rho_0^{-1} \int_{\mathbf{B} \cap \bar{\mathbf{B}}_{\rho_0}(0, 0, \pm 1)} |\nabla u|^2 dx$$

are small. One finds as before that u then has small oscillation on $\bar{\mathbf{B}} \cap \partial\mathbf{B}_\rho(0, 0, \pm 1)$ for a suitable $\rho \in (\frac{1}{2}\rho_0, \rho_0)$. Thus u maps $\partial(\mathbf{B} \cap \bar{\mathbf{B}}_\rho(0, 0, \pm 1))$ close to the pole $u(0, 0, \pm 1)$. Minimality again implies that the entire polar cap $\bar{\mathbf{B}} \cap \bar{\mathbf{B}}_\rho(0, 0, \pm 1)$ is mapped by u into the corresponding open hemisphere and Hölder continuity on the cap follows. $\square$

4.3 Lemma (tangent maps). *The only homogeneous-degree-0 axially symmetric maps from $\mathbf{B}$ to $\mathbf{S}^2$ that minimize energy among axially symmetric maps are $+\Lambda$, $-\Lambda$, $+\Omega$, and $-\Omega$ where*

$$\Lambda(x) = (x_1, x_2, x_3)/|x| \quad \text{and} \quad \Omega(x) = (x_1, x_2, -x_3)|x|.$$

Proof. Recall that any finite energy harmonic map v from $\mathbf{S}^2$ to $\mathbf{S}^2$ determines, via stereographic projection, a function $f_v: \mathbb{C} \rightarrow \mathbb{C}$, which is a rational function of z or $\bar{z}$. Assuming also that v is axially symmetric, we infer from (2.1) with $n = 1$ that f_v maps $\{0, \infty\}$ to $\{0, \infty\}$ and the real axis to itself. Moreover, the restriction of f_v to a circle centered at the origin is simply multiplication by a real number. These facts imply that f_v equals either λz or $\lambda/\bar{z}$ with $0 \neq \lambda \in \mathbb{R}$. Finally, by computing the energy of the homogeneous extension from different points on the Z -axis as in [BCL] or [HL, 5.7], we infer from energy minimality that $|\lambda| = 1$. The four remaining functions $+z$, $-z$, $+1/\bar{z}$, and $-1/\bar{z}$ correspond to respectively $+\Lambda$, Ω , $-\Lambda$, and $+\Omega$. Being rotations of $x/|x|$, these four minimize energy among all maps, in particular among axially symmetric maps, having the same boundary data [BCL]. $\square$

4.3 Theorem. *There is a positive $\alpha < 1$ so that if $u: \mathbf{B} \rightarrow \mathbf{S}^2$ is energy minimizing among axially symmetric maps and a is a singular point of u in $\mathbf{B}$, then $|u(x) - u_0(x - a)| \leq C|x - a|^\alpha$ where u_0 is either $+\Lambda$, $-\Lambda$, $+\Omega$, or $-\Omega$.*

Proof. As observed in the proofs of 4.1 and 2.1, a must lie on the Z -axis and any tangent map u_0 of u at a must be axially symmetric. The

strong convergence implies as in [SU₁, 5.2] that u_0 must also minimize energy among axially symmetric maps. The uniqueness of this tangent map and the decay estimate follow by repeating arguments of L. Simon [S] and R. Gulliver and B. White [GW]. Here the Jacobi fields are axially symmetric and integrable via axially symmetric maps.

5. The Number of Singularities in Minimizers

By the singularity classification of [BCL], an energy-minimizing map from $\mathbf{B}$ to $\mathbf{S}^2$ whose boundary data has positive degree k will necessarily have at least k singular points. In [HL₁] we constructed degree-0 examples of boundary data whose energy minimizers also must have singularities. Here we give some examples of minimizers for which the number of singularities is precisely k .

First we will sketch one construction which involves a dumbbell-shaped domain. This has been worked out independently by Libin Mou. Let $\Omega_\epsilon = B_- \cup H_\epsilon \cup B_+$ where

$$B_\pm = \mathbf{B}_1((0, 0, \pm 2)), \quad H_\epsilon = \{x \in \mathbb{R}^3: x_1^2 + x_2^2 < \epsilon^2, -1 < x_3 < 1\}.$$

Analogous with [HL₁], we choose uniformly Lipschitz boundary data $g_\epsilon \in C^\infty(\partial\Omega_\epsilon, \mathbf{S}^2)$ so that

$$\begin{aligned} g_\epsilon(x) &= \pm(x - (0, 0, \pm 2))/|x - (0, 0, \pm 2)| \quad \text{on } \partial B_\pm \sim B_{2\epsilon}((0, 0, \pm 1)), \\ g_\epsilon(x) &= (0, 0, -1) \quad \text{on } \partial H_\epsilon \sim (B_- \cup B_+). \end{aligned}$$

Let u_ϵ be an energy minimizing map with $u_\epsilon|_{\partial\Omega_\epsilon} = g_\epsilon$. Thanks to the uniqueness [BCL] of the energy minimizer $x/|x|$ and the strong convergence of minimizers, we see that $u_\epsilon|_{B_\pm}$ converges strongly in H^1 to $\pm(x - (0, 0, \pm 2))/|x - (0, 0, \pm 2)|$. Using [HL₃] or [AL], including the uniform boundary regularity lemma, we see that, for ϵ sufficiently small,

$$u_\epsilon|_{(B_\pm \sim B_{1/8}(0, 0, \pm 1))}$$

has precisely one singularity. Concerning the remaining middle region of Ω_ϵ , we note that the strong convergence also implies that, for ϵ sufficiently small, u_ϵ maps the small spherical cap $B_\pm \cap \partial B_{1/4}((0, 0, \pm 1))$ near the South Pole $(0, 0, -1)$. For such ϵ , u_ϵ thus maps the entire boundary of the small dumbbell

$$D_\epsilon = B_{1/4}((0, 0, -1)) \cup H_\epsilon \cup B_{1/4}((0, 0, 1))$$

into the open Southern Hemisphere. By minimality $u_\epsilon \mid D_\epsilon$ has image in this hemisphere and so [HKW] is free of singularities. We conclude that u_ϵ has precisely 2 singularities.

Adding additional balls one readily obtains examples with an arbitrary number of singularities. Libin Mou has begun making such a bridge construction using nonminimizing configurations involving higher degree singularities by modifying arguments of N. Smale [Sm] for minimal hypersurfaces.

Finally we give an axially symmetric construction which will illustrate some of the differences between minimizers among all maps and minimizers in the axially symmetric class. For convenience we will work with the fixed cylindrical domain

$$U = \{x \in \mathbb{R}^3: x_1^2 + x_2^2 < 1, -1 < x_3 < 1\}$$

and the corresponding orbit space $E = \{(r, z) \in \mathbb{R}^2: 0 \leq r < 1, -1 < z < 1\}$.

5.1 Theorem. *Suppose $k \in \{1, 2, \dots\}$ and $v: \partial U \rightarrow \mathbb{S}^2$ is a Lipschitz axially symmetric mapping which corresponds (as in §3) to a function $g: (\partial E) \cap \{r > 0\} \rightarrow \mathbb{R}$ for which*

$$g(r, 1) \equiv 0, \quad g(r, -1) \equiv k\pi, \quad \text{and } g(1, t) \text{ is monotonic on } \{-1 \leq t \leq 1\}.$$

Then the singular set of a mapping u that minimizes energy among axially symmetric maps with trace v has precisely k singular points.

Proof. Suppose $u = u_\varphi$ with $\varphi: E \rightarrow \mathbb{R}$, as in §3. By considering the one dimensional monotone rearrangement, in the Z direction, as discussed by B. Kawohl [Ka, Th. 2.13], we deduce that $\varphi(r, \cdot)$ is monotonic on $\{-1 \leq t \leq 1\}$ for every positive $r < 1$. Moreover, by the regularity of φ away from $\{r = 0\}$, the total variation of $\varphi(r, \cdot)$ on $\{-1 \leq t \leq 1\}$ remains $k\pi$ as $r \downarrow 0$. From 4.3 and 4.2 we conclude that u has precisely k singularities, whose degrees alternate between 1 and -1 as one descends the Z -axis. $\square$

5.2 Corollary. *For k sufficiently large, a mapping w that minimizes energy among all maps having trace v , as in 5.1, is **not** axially symmetric. Moreover, $\{R_{-\theta} \circ w \circ R_\theta: -\varepsilon \leq \theta \leq \varepsilon\}$ forms, for ε sufficiently small, a one parameter family of distinct energy-minimizing maps each having trace v .*

Proof. By the uniform boundary regularity theorem [HL₃, 2.1],

$$(\text{Sing } w) \cap \mathbb{B}_\delta((0, 0, \pm 1)) = \emptyset$$

for some $\delta > 0$, independent of k because v is constant on neighborhoods of $(0, 0, 1)$ and $(0, 0, -1)$ that are independent of k . Finally by the interior bounds of [HKL₂], [AL] or [HL₃, §4],

$$\text{card}\{a \in \text{Sing } w : \text{dist}(a, \partial U) > \delta\} < N_\delta$$

for some constant N_δ depending only on δ (actually $N_\delta = c/\delta$).

For $k \geq N_\delta$ we conclude that some singularity must lie off of the Z -axis and hence, as in 2.1, w is not axially symmetric, and the equation $w \circ R_\theta \equiv R_\theta \circ w$ holds for at most a finite number of angles $\theta \in [0, 2\pi)$.

6. Final Remarks

6.1 Caccioppoli Inequality. Unlike a minimizer in the full class of mappings, a minimizer in the axially symmetric subclass does not satisfy the “Caccioppoli-type” estimate of [HL₂, 6.2]. In fact, such an estimate would lead, as in [HKL₂], [AL] or [HL₃, §4], to energy bounds, uniform boundary regularity, and estimates on the total number of singularities, which would, as in 5.2, contradict the construction of 5.1.

6.2 The dumbell. In [HL₁], we obtained degree-0 boundary data for which the energy minimizer has an arbitrary positive even number of singularities. This involved using translations of the singular maps $+x/|x|$ and $-x/|x|$, and here the axially symmetric boundary data was monotone in the sense of 5.1. If we replace $-x/|x|$ by the other axially symmetric degree -1 singular tangent map $\Omega(x) = (x_1, x_2, -x_3)/|x|$, then we can achieve the same conclusion but now with axially symmetric boundary data for which the corresponding g , as in §3, satisfies $|g| \leq \pi$. We can also get $|g| < \pi - \delta$ for some positive δ at the expense of adding some small energy in the large middle region. With Zhang’s theorem we thus have, as in [HL₃, §5], a specific example of boundary data for which there is one regular and one singular harmonic map.

6.3 n -axially symmetric maps. The results of §3–§5 easily carry over for n -axially symmetric maps. In the formula for $E(\varphi)$ one should add the factor n^2 to the $r^{-1} \sin^2 \varphi$ term. Then $\frac{1}{2}$ should be replaced by $\frac{1}{2}n^2$ in (3.1). This does not substantially change the proofs of 3.1, 3.2, 4.1, or 4.2 as the types of the P.D.E.’s and O.D.E.’s considered are unchanged. In 4.3, the classification of harmonic maps from S^2 to S^2 and the n -axially symmetric constraint now imply that the tangent map at a singularity must correspond to either z^n , $-z^n$, $\bar{z}^{-n}$, or $-\bar{z}^{-n}$. These new tangent maps should be used in the statement and proof of 4.3 and in the constructions of §5. In 5.1

note that the boundary map v was degree 1 if k was odd and degree 0 if k was even. Now it is either degree n or degree 0.

REFERENCES

- [AL] F. J. Almgren, Jr. and E. Lieb, *Singularities of energy minimizing maps from the ball to the sphere: Examples, counterexamples, and bounds*, Ann. Math. **128** (1988), 483–530.
- [B] A. Baldes, *Stability and uniqueness properties of the equator map from a ball into an ellipsoid*, Math. Z. **185** (1984), 505–516.
- [BCL] H. Brezis, J.-M. Coron, and E. Lieb, *Harmonic maps with defects*, Comm. Math. Physics **107** (1986), 649–705.
- [C] R. Courant, *Dirichlet's principle*, Interscience, 1950.
- [CH] J.-M. Coron and F. Helein, *Harmonic diffeomorphisms, minimizing harmonic maps, and rotational symmetry*, preprint, Univ. Paris Sud, Orsay, 1988.
- [CHKLL] R. Cohen, R. Hardt, D. Kinderlehrer, S.Y. Lin, and M. Luskin, *Minimum energy configurations for liquid crystals; computational results*, in *Theory and Applications of Liquid Crystals*, IMA, **5**, Springer, 1986.
- [D] Ding, *Symmetric harmonic maps between spheres*, preprint.
- [E] J. Ericksen, *Static theory of point defects in nematic liquid crystals*, preprint.
- [GW] R. Gulliver and B. White, *The rate of convergence of a harmonic map at a singular point*, Math. Ann. **283** (1989), 539–550.
- [Ha] R. Hardt, *Point and line singularities in liquid crystals*, this volume.
- [He] F. Helein, *Minima de la fonctionnelle énérgie libre des cristaux liquides*, Calcul des Variations, to appear in C.R.A.S.
- [HK₁] R. Hardt and D. Kinderlehrer, *Mathematical questions of liquid crystal theory*, in *Theory and Applications of Liquid Crystals*, IMA, **5**, Springer, 1986.
- [HK₂] R. Hardt and D. Kinderlehrer, *Mathematical questions of liquid crystal theory*, in College de France Seminar, vol. IV(3), 1987.
- [HKL₁] R. Hardt, D. Kinderlehrer, and F.H. Lin, *Existence and partial regularity of static liquid crystal configurations*, Comm. Math. Physics **105** (1986), 547–570.
- [HKL₂] R. Hardt, D. Kinderlehrer, and F.H. Lin, *Stable defects of minimizers of constrained variational problems*, Ann. Inst. Henri Poincare, Analyse non lineaire, **5**, no. 4 (1988), 297–322.
- [HKLu] R. Hardt, D. Kinderlehrer, and M. Luskin, *Remarks about the mathematical theory of liquid crystals*, in *Calculus of Variations and*

- Partial Differential Equations*, S. Hildebrandt, D. Kinderlehrer, and M. Miranda, ed., Lecture Notes in Math. **1340**, 123–138.
- [HKW] S. Hildebrandt, H. Kaul, and K. Widman, *An existence theory for harmonic maps of Riemannian manifolds*, Acta Math. **188** (1977), 1–16.
- [HL₁] R. Hardt and F.H. Lin, *A remark on H^1 mappings*, Manuscripta Math. **56** (1986), 1–10.
- [HL₂] R. Hardt and F.H. Lin, *Mappings minimizing the L^p norm of the gradient*, Comm. Pure & Appl. Math. **40** (1987), 555–588.
- [HL₃] R. Hardt, and F.H. Lin, *Stability of singularities of minimizing harmonic maps*, to appear in J. Diff. Geom. **28** (1988).
- [JK₁] W. Jager and H. Kaul, *Uniqueness and stability of harmonic maps and their Jacobi fields*, Manuscripta Math. **28**(4) (1979), 269–271.
- [JK₂] W. Jager and H. Kaul, *Rotationally symmetric harmonic maps from a ball into a sphere and the regularity problem for weak solutions to an elliptic systems*, J. Reine Angew. Math. **343** (1983), 146–161.
- [K] D. Kinderlehrer, *Recent developments in liquid crystal theory*, Proceedings of Colloq. Lions.
- [Ka] B. Kawohl, *Rearrangements and convexity of level sets in PDE*, Lecture Notes in Math. **1150**, Springer-Verlag, Berlin-Heidelberg-New York, 1985.
- [La] H.B. Lawson, *The equivariant Plateau problem*, Trans. A.M.S. **173** (1972), 231–250.
- [LSY] Lin, San-Yih, *Numerical analysis of liquid crystal problems*, Thesis, U. of Minnesota, 1987.
- [LFH] Lin, Fang-hua, *Nonlinear theory of defects in nematic liquid crystals —phase transition and flow phenomena*, to appear in Comm. Pures & Appl. Math.
- [LM] J.L. Lions and E. Magenes, *Nonhomogeneous boundary value problems and applications I*, Springer-Verlag, Berlin-Heidelberg-New York, 1962.
- [M₁] F. Morgan, *A smooth curve in $\mathbb{R}^3$ bounding a continuum of minimal surfaces*, Arch. Rat. Mech. Anal. **75** (1981), 193–197.
- [M₂] F. Morgan, *Finiteness of the number of stable minimal hypersurfaces with a fixed boundary*, Indiana U. Math. J. **35**(4) (1986), 779–833.
- [S] L. Simon, *Isolated singularities of extrema of geometric variational problems*, Lecture Notes in Math. **1186**, Springer-Verlag, Berlin-Heidelberg-New York, 1986.
- [SaU] J. Sacks and K. Uhlenbeck, *The existence of minimal immersions of 2-spheres*, Ann. of Math. **113** (1981), 1–24.
- [Sm] R.T. Smith, *Harmonic maps of spheres*, Amer. J. Math. **97**(2) (1975), 364–385.

- [Sma] N. Smale, *Minimal hypersurfaces with many isolated singularities*, Ann. Math. **130** (1989), 603–642.
- [St] E. Stein, *Singular integrals and differentiability properties of functions*, Princeton, 1970.
- [SU₁] R. Schoen and K. Uhlenbeck, *A regularity theory for harmonic maps*, J. Diff. Geom. **17** (1982), 307–335.
- [SU₂] R. Schoen and K. Uhlenbeck, *Boundary regularity and the Dirichlet problem of harmonic maps*, J. Diff. Geom. **18** (1983), 253–268.
- [SU₃] R. Schoen and K. Uhlenbeck, *Regularity of minimizing harmonic maps into the sphere*, Inventiones Math. **78** (1984), 89–100.
- [Z] D. Zhang, *Axially symmetric harmonic maps*, preprint, Univ. Cal. San Diego, March, 1987.

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